

A Kernel-Checked Asymptotic Formalization for Erdős Problem 848

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Abstract

For $N \geq 1$, let $A \subseteq \{1, \dots, N\}$ and suppose that $ab + 1$ is not squarefree for every $a, b \in A$. Sawhney proved that, for all sufficiently large N , the largest possible cardinality of A is the number of integers congruent to 7 modulo 25. We give a Lean 4 formalization of this asymptotic theorem and a short paper-to-kernel map. The literal interval statement is obtained from the formalization's zero-based range convention by a kernel-checked indexing lemma. We also kernel-check two exact small instances, the exact online-colouring trace through $N = 5,000,000$, and the precise finite-reduction statement.

This note does not claim a complete all- N resolution of Erdős Problem 848; the remaining tail $N \geq 5,000,000$ is not yet closed. Every numbered mathematical result asserted here is mapped to a declaration whose Lean axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

1 The problem and the sharp constructions

An integer is squarefree when no prime square divides it. Write

$$[N] = \{1, \dots, N\}, \quad C_r(N) = \{n \in [N] : n \equiv r \pmod{25}\}.$$

Definition 1.1. A finite set $A \subseteq [N]$ is *848-admissible* if

$$ab + 1 \text{ is not squarefree for every } a, b \in A.$$

Let $P(N)$ denote the statement that every 848-admissible $A \subseteq [N]$ satisfies $|A| \leq |C_7(N)|$.

In Lean these are `NonSquarefreeProductProp` and `OriginalProblem848Statement`. Repeated elements are allowed in the quantified pair, so the diagonal condition $a^2 + 1$ is part of the hypothesis.

Proposition 1.2. *Both $C_7(N)$ and $C_{18}(N)$ are 848-admissible. Moreover $|C_{18}(N)| \leq |C_7(N)|$.*

Proof. If a, b lie in the same one of the two displayed classes, then

$$ab + 1 \equiv 7^2 + 1 \equiv 18^2 + 1 \equiv 0 \pmod{25}.$$

Thus $5^2 \mid ab + 1$. The cardinality comparison is the elementary count of the two residue classes in $[N]$. The formal construction theorems are `A7_has_property_public` and `A18_has_property_public`; the exact range counts are `A7_card_public`, `A18_card_public`, and `A18_card_le_A7_public`. □

The lower bound $|A| \geq |C_7(N)|$ is therefore immediate. The content of the problem is the matching upper bound.

2 The stability theorem

The formalization follows Sawhney’s stability route [1]. It first uses the diagonal condition to constrain each element, then uses off-diagonal products to rule out a dense mixture of incompatible residue patterns.

For compatibility with Mathlib, the large source file initially works in `Finset.range N`, namely $\{0, \dots, N - 1\}$. Put

$$A_7(N) = \{n < N : n \equiv 7 \pmod{25}\}, \quad A_{18}(N) = \{n < N : n \equiv 18 \pmod{25}\}.$$

Theorem 2.1 (Formalized stability theorem). *There exist $\eta > 0$ and $N_0 \in \mathbb{N}$ such that $\eta < 1/25$ and, for every $N \geq N_0$, every 848-admissible $A \subseteq \{0, \dots, N - 1\}$ satisfying*

$$|A| \geq (1/25 - \eta)N$$

is contained in either $A_7(N)$ or $A_{18}(N)$.

Proof. This is the proposition `SawhneyMainAt`, instantiated by the kernel-checked theorem `sawhney_main`. The formal proof chooses $\eta = 1/2000$ and a non-explicit threshold large enough for its prime-counting error. Its main ingredients are as follows.

First, if $p^2 \mid a^2 + 1$, then $p \equiv 1 \pmod{4}$, and the congruence has two roots modulo p^2 . This supplies the diagonal sieve. Second, mixed 7- and 18-classes force a prime-square witness other than 5^2 in their cross-products. Counts in fixed classes modulo $25p^2$, $50p^2$, and $100p^2$ give the off-diagonal sieve. Third, the proof evaluates the finite reciprocal-square contribution through the fixed cutoff 2000 and bounds the remaining tail. A prime-counting $o(N)$ estimate absorbs the additive errors after increasing N_0 . Finally, an exhaustive parity and residue case split shows that any set above the stated density threshold is contained in one of the two base classes.

These steps are declarations in `Asymptotic.lean`; no numerical output is imported as an axiom or trusted certificate. $\square$

3 The asymptotic endpoint

Theorem 3.1 (Range-indexed endpoint). *There exists N_0 such that, for every $N \geq N_0$ and every 848-admissible $A \subseteq \{0, \dots, N - 1\}$,*

$$|A| \leq |A_7(N)|.$$

Proof. Suppose $|A| > |A_7(N)|$. The exact residue-class count implies $|A| \geq (1/25 - \eta)N$. Theorem 2.1 then puts A inside $A_7(N)$ or $A_{18}(N)$. In the first case cardinality is immediate; in the second it follows from $|A_{18}(N)| \leq |A_7(N)|$. In Lean this deduction is `problem_848_large_of_sawhney`, and the unconditional existential endpoint is `problem_848_asymptotic`. The fixed- N proposition is `Problem848Statement`. $\square$

The zero-based convention must be bridged explicitly rather than dismissed as asymptotically harmless.

Lemma 3.2 (Index bridge). *For every N ,*

$$C_7(N) = A_7(N + 1).$$

Consequently the range-indexed statement at $N + 1$ implies $P(N)$.

Proof. The two sets have the same upper bound. Their only possible difference is 0, which is not congruent to 7 modulo 25. The corresponding Lean declarations are `originalA7_eq_range` and `original_of_range_succ`. $\square$

Theorem 3.3 (Literal asymptotic endpoint). *There exists N_0 such that $P(N)$ holds for every $N \geq N_0$. Equivalently, for all sufficiently large N , the exact maximum is $|C_7(N)|$.*

Proof. Apply Theorem 3.1 at $N + 1$ and then Lemma 3.2. The construction in Proposition 1.2 supplies equality. This is `erdos848_original_asymptotic`. $\square$

4 Finite reduction and the certified prefix

The formal source also contains direct finite checks at range sizes 50 and 100. After applying the index bridge, they become the following literal instances.

Proposition 4.1. *The statements $P(49)$ and $P(99)$ hold.*

Proof. The kernel-checked declarations are `erdos848_original_N49` and `erdos848_original_N99`. Their upstream range-indexed proofs reduce the diagonal candidates to explicit finite sets and exclude oversized admissible subsets by `decide`; they do not use `native_decide`. $\square$

Corollary 4.2 (Exact finite reduction). *There exists an integer N_0 such that $P(N)$ holds for every $N \geq N_0$ and*

$$(\forall N, P(N)) \iff (\forall N < N_0, P(N)).$$

Proof. Choose N_0 from Theorem 3.3. The forward direction is restriction. Conversely, use the assumed finite prefix below N_0 and the asymptotic theorem at or above N_0 . The Lean declaration is `erdos848_finite_reduction`. $\square$

Proposition 4.3 (Online-colouring soundness). *Fix N . Suppose that every $x \in [N]$ for which $x^2 + 1$ is nonsquarefree is assigned a colour $c_N(x) \in C_7(N)$, and suppose that distinct diagonal candidates of the same colour satisfy that $xy + 1$ is squarefree. Then $P(N)$ holds. Consequently, a certificate providing such a colouring separately at every prefix through L proves $P(N)$ for every $1 \leq N \leq L$.*

Proof. Restrict c_N to an 848-admissible set A . Two distinct members of A cannot have the same colour, since their cross-product plus one is required to be nonsquarefree. Thus $c_N|_A$ is an injection into $C_7(N)$. The Lean interface is `PrefixColouringState`; its soundness declarations are `originalProblem_of_prefixColouringState` and `originalProblem_prefix_of_colouringCertificate`. $\square$

Theorem 4.4 (Kernel-checked prefix). *The statement $P(N)$ holds for every $N \leq 5,000,000$.*

Proof. The official payload records 525,808 diagonal vertices: 200,000 residue-7 anchors, 200,000 residue-18 candidates, and 125,808 outside-base candidates. It records the interval history of every anchor and a covering reverse-assignment history for every non-anchor candidate. Each assignment identifies one of the three occupant roles at an anchor. The checker verifies that the histories cover every prefix, that equal-colour distinct occupants have a squarefree cross-product plus one, and that every nonsquarefree diagonal candidate occurs in the indexed sequence. Primality and squarefreeness facts are discharged by the split five-million Pratt and factor certificates. Applying Proposition 4.3 yields the Lean theorem `erdos848_through_five_million`. The older range

endpoints `erdos848_prefix_close`, `erdos848_first_low_close`, and `erdos848_second_low_close` are direct projections of this single certificate, not additional assumptions. $\square$

Proposition 4.5 (Exact remaining-tail assembly). *Suppose $P(N)$ is proved for every $N \geq 5,000,000$. Then $P(N)$ holds for every N .*

Proof. Use Theorem 4.4 below the cut and the assumed tail theorem at or above it. Lean checks this literal assembly as `erdos848_full_of_five_million_tail`. This proposition does not assert the tail premise; it isolates the only remaining all- N obligation. $\square$

Corollary 4.2 is a logical reduction, not by itself a completed finite computation. The threshold produced inside the formal proof is non-explicit because it is chosen after an eventual prime-counting bound. The current public problem record accordingly describes Problem 848 as “resolved up to a finite check” [2]. The explicit range through five million is now certified by Theorem 4.4. No unchecked exploratory finite-search output is used by any theorem in this repository.

The earlier diagonal and Hall reductions are retained as independently auditable structural layers. In particular, the generated diagonal certificate proves that every outside-base $x \leq 5,000,000$ with nonsquarefree $x^2 + 1$ occurs in an exact sorted marker, checks its odd and two odd mod-4 projections, and proves the rational count envelopes consumed by the Hall argument. The kernel endpoints include

```
GeneratedLowDiagonalCoverage.certificateContainsOfNotSquarefree
GeneratedLowDiagonalCoverage.firstLowDiagonalBounds
GeneratedLowDiagonalCoverage.secondLowDiagonalBounds
```

These layers are no longer open premises for the published finite range: the single exact trace certificate proves the complete statement through five million directly.

Remark 4.6 (Boundary of the earlier all- N claim). An earlier exploratory draft assembled an all- N claim from an online finite colouring and several analytic block programs. The complete online colouring through 5,000,000 has now been lifted to the kernel-checked payload used in Theorem 4.4. The final $N \geq 5,000,000$ argument remains a separate obligation. No earlier program output is treated as a proof of that remaining statement.

5 Paper-to-kernel map and audit boundary

Paper label	Lean declaration(s)	Role
def:admissible	OriginalProblem848Statement	literal statement
prop:base-constructions thm:stability	A7_has_property_public, A18_has_property_public sawhney_main	lower bound structural theorem
thm:range-asymptotic	problem_848_asymptotic	range endpoint
lem:index-bridge thm:original-asymptotic	original_of_range_succ erdos848_original_asymptotic	indexing main result
prop:small-instances cor:finite-reduction	erdos848_original_N49, erdos848_original_N99 erdos848_finite_reduction	exact checks remaining boundary
Proposition 4.3	originalProblem_of_prefixColouringState	certificate interface
thm:prefix-close Proposition 4.5	erdos848_through_five_million erdos848_full_of_five_million_tail	exact prefix exact assembly

The machine-readable version of this table is `paper/theorem-map.json`. Its checker requires each label to occur in the manuscript, each declaration to occur in the publication Lean closure, and each declaration to be named by a `#check` command. A separate `--trust=0` file prints the axioms of the publication endpoints. The exact output for each is

[propxt, Classical.choice, Quot.sound],

the standard classical and quotient dependencies used through Mathlib.

The principal formal source was developed by Raymond Jung and contributors and released under Apache 2.0 [3]. This repository preserves the source file and its contributor header, pins its SHA-256, and adds only the literal-interval bridge and publication audit surface. The author used OpenAI Codex to reorganize the public package, draft the bridge declarations and verification scripts, execute the builds and audits, and edit this note. All numbered claims remain subject to the Lean kernel checks described above.

6 Conclusion

The public mathematical closure established here consists of the asymptotic theorem and the exact range $N \leq 5,000,000$. The conjectured $7 \pmod{25}$ cardinality is therefore kernel-checked on that prefix and for every sufficiently large N . A complete all- N solution still requires a kernel proof of the final tail in Proposition 4.5; until that premise is supplied, this work should not be reported as a full resolution of Erdős Problem 848.

References

- [1] M. Sawhney, *On $A \subseteq [N]$ such that $ab + 1$ is never squarefree for $a, b \in A$* , 2025. https://www.math.columbia.edu/~msawhney/Problem_848.pdf

- [2] T. F. Bloom, *Erdős Problem 848*, Erdős Problems. <https://www.erdosproblems.com/848>
- [3] R. Jung and contributors, *Lean 4 formalization of the asymptotic result for Erdős Problem 848*, erdos-banger, commit 48e9c1aeb13a6e075d78ecf42dc1f2839d5ff071, 2026. <https://github.com/The-Obstacle-Is-The-Way/erdos-banger>